

Tax Productivity and Development: A Quantitative Macroeconomic Analysis

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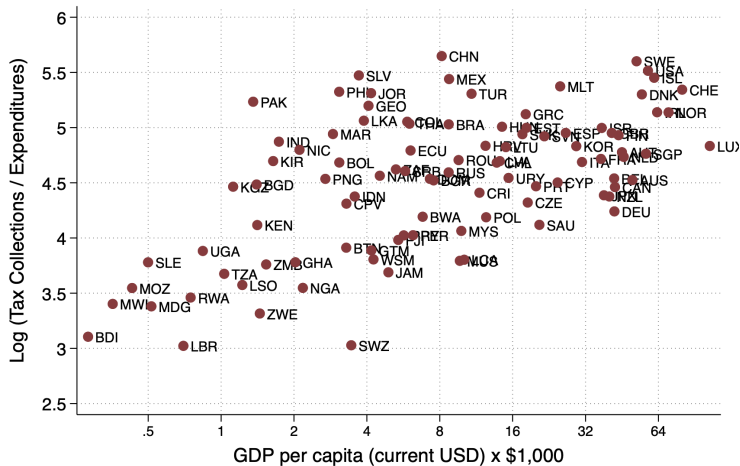
Motivation

- **Tax collection:** core task of government; prerequisite for any public spending.
- Central to '**state capacity**' theory of development,
Besley and Persson (2011), Besley, Ilzetzki, and Persson (2013),
Mayshar, Moav, and Pascali (2022), Tilly (1990), others.
- Of taxes in *The Wealth of Nations*
"Every tax ought to be so contrived as both to take out and to keep out of the pockets of the people as little as possible over and above what it brings into the public treasury of the state. . . the cost of collecting it may be very considerable, and in the worst cases may amount to more than it brings in."- Adam Smith

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- Tax collection capacity is not easy to measure,
 - Typically proxied by **Taxes/GDP**, but quite indirect measure
 - Theories of state capacity $\Rightarrow$ tax capacity increases Taxes and GDP.
- Little consensus on **quantitative importance** of tax collection capacity,
 - **Q:** How much long-run GDP per capita increases if tax collection capacity in LICs raised?

1. Tax Collection Capacity Increases with GDP per Capita



Overview of This Paper

- **Data:** new data on tax collection across 100 countries
 - Tax collections (T) / collection expenditures (X)
 - Document strong positive correlation btw T/X and GDP per capita over:
 - (1) cross-section, (2) time series, and (3) across local states
- **Stylized model** of tax productivity and development
 - Role of marginal and average product of tax collection
- **Quantitative analysis:** dynamic G.E. model to interpret data + simulate counterfactuals
 - A **Tax administration** in workhorse Ramsey model
 - B **Public capital** complementary to private capital à la Barro (1990), Baxter-King (1993)
 - C Reallocation to and from modern/traditional sector

Key Results

1. Tax collection capacity increases with GDP per capita
2. Decreasing returns in tax collection can generate differences in long-run GDP
3. We provide Ramsey model (& novel formulas) for optimal state capacity
4. Quantitatively, GDP gains from tax productivity shocks are an order of magnitude larger than predicted by sectoral share accounting ("Hulten").

2. DRS in Tax Collection Can Generate Differences in LR GDP

Environment

- Production linear in labor, normalized to 1, $Y = L = 1$.
- Tax rate $\tau \in [0, 1]$ finances a public good G .
- *Average* cost of raising one dollar of revenue $\mu(\tau)$.
- *Marginal cost of public funds (MCPF)* $\mu'(\tau)$.
- Consumption C , public goods G , and preferences U :

$$C = (1 - \mu(\tau)\tau)Y, \quad G = \tau Y \quad U(C, G) = C^\beta G^\alpha, \quad \beta, \alpha > 0.$$

- The planner maximizes

$$\max_{\tau \in [0, 1]} (1 - \mu(\tau)\tau)^\beta (\tau)^\alpha. \quad (1)$$

- Micro-found link between the tax collection technology and the cost of public funds

Constant returns in tax collection

- Tax administrators collect $AL_C = \tau L_Y$, $A > 0$ is tax productivity, $L_Y + L_C = L = 1$.
- Consumption public goods are $C = (1 - \tau)L_Y$, $G = \tau L_Y$.
- Solving for L_Y , planner's problem equiv to (1) with constant aver. cost of public funds μ :

$$\mu \approx 1 + \frac{\alpha + \beta}{\beta} \frac{1}{A}. \quad (2)$$

- optimal amount of taxed collected in (1) is $\tau = \frac{\alpha}{\mu(\alpha + \beta)}$.
- If μ is close to 1, taxes collected close to frictionless tax collection, $\mu = 1$ and $\tau^* = \frac{\alpha}{(\alpha + \beta)}$
- With $A = 30$ (average T/X_{LIC}), $\beta = 2/3$, $\alpha = 1 - \beta$, $\mu = 1.05$, and $G = 0.31$.
- For high-income countries, $A \approx 200$, $\mu = 1.015$, i.e. $G = 0.33$.
- CRS tax collection can't generate meaningful differences in development across countries.

Decreasing returns in tax collection

- Suppose now $AL_C^\eta = \tau L_Y$, where $0 < \eta < 1$.
- Labor market clearing $L_Y + (\frac{\tau L_Y}{A})^{1/\eta} = 1 \implies$ variable cost of funds $\mu(\tau)$ with $\mu'(\tau) > 0$.
- FOC of planner choice of τ in (1)

$$\tau^2 = \frac{\alpha - \tau\mu(\tau)(\alpha + \beta)}{\beta\mu'(\tau)}. \quad (3)$$

- numerator is of the same order of magnitude across countries
- But, small differences in the MCPFs can lead to meaningful differences in taxes collected and public goods.

3. A Ramsey Model of Tax Collection Capacity and Development

Overview of Model Environment

- Three novel model features to quantify workhorse Ramsey model

Judd (1985), Chamley (1986), Straub and Werning (2020), Auclert, Cai, Rognlie, and Straub (2024)

1. Tax administration in workhorse Ramsey model

- One input: tax collectors L_C
- Output: tax rate on modern sector wages, $T_t = G(A^C, L_{C,t})w_{M,t}L_{M,t}$
- Tax productivity A^C is exogenous, captures “tax collection capacity”

2. Public capital complimentary to private capital: $Y_M = F(A^M, k_G, k, L_M)$,

Musgrave (1959), Barro (1990), Baxter-King (1993), Weinzeirl (2018)
World Bank Long-Run Growth Model accounting for corruption

3. Reallocation to and from modern/traditional sector

Technology: Modern and Traditional Sectors

- **Modern Sector:** production function with public capital k_G , **taxed wages:**

$$Y_M = F(A^M, k_G, k, L_M),$$

hire private (wage $w_{M,t}$) and rent price capital (rate r_t)

$$\max_{k_t, L_{M,t}} F(A^M, k_{G,t}, k_t, L_{M,t}) - w_{M,t} L_{M,t} - r_t k_t.$$

- **Traditional Sector:** linear production function, **untaxed:**

$$Y_T = A^T L_T,$$

wage $w_T = A^T$.

Public Sector and Government

- **Tax Collection:** Taxes T_t are collected on modern sectors wages at rate $G(A^C, L_{C,t}) \leq 1$,

$$T_t = G(A^C, L_{C,t})w_{M,t}L_{M,t}.$$

- **Government budget:** can borrow to finance budget.

$$w_{M,t}L_{C,t} + R_t b_t + k_{t+1}^G \leq T_t + b_{t+1} + (1 - \delta^G)k_t^G + \omega_t(1 + r_t - \delta)k_t.$$

Public capital evolves as:

$$k_{G,t+1} = (1 - \delta^G) k_{G,t} + T_t.$$

Government cannot borrow to finance public capital.

Households

- Modern and traditional sector workers form a **representative household**

$$\max_{\{c_t, k_{t+1}, b_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t, L_{M,t}, L_{T,t})$$

- **Budget constraint**

$$c_t + k_{t+1} + b_{t+1} \leq (1 - \omega_t)(1 + r_t - \delta)k_t + (1 - \underbrace{G(A^C, L_{C,t})}_{\text{tax rate}})w_{M,t}L_{M,t} + w_T L_{T,t} + R_t b_t.$$

Initial private capital $k_0 = \bar{k}_0$.

Equilibrium

A competitive equilibrium is a set of allocations

$$\{c_t, L_{T,t}, L_{M,t}, L_{C,t}, k_t, k_{G,t}, b_t\}$$

and prices

$$\{w_{T,t}, w_{M,t}, r_t, R_t\},$$

such that:

1. Firm Optimality:

- Modern firms take prices and $k_{G,t}$ as given, and maximize profit.
- Traditional firms hire labor at wage $w_T = A^T$.

2. Household Optimality: The household maximizes utility for given prices and L_C .

3. Budget constraints: Government BC and public capital accumulation equations hold.

4. Market Clearing: The goods market clears:

$$c_t + k_{t+1} + \underbrace{w_{M,t} L_{C,t}}_{\text{collection personnel wages}} + k_{G,t+1} \leq (1 + r_t - \delta) k_t + w_{M,t} L_{M,t} + A^T L_{T,t} + (1 - \delta^G) k_{G,t}.$$

Planner's Problem

- Government chooses the sequence of $\pi = \{L_{C,t}, \omega_t\}_{t=0,\dots,\infty}$ to maximize welfare:

$$\max_{\pi} \sum_{t=0}^{\infty} \beta^t u(c_t(\pi), L_{M,t}(\pi), L_{T,t}(\pi)),$$

subject to:

- the aggregate resource constraint,
- an **implementability condition** for consistency with household optimization,
- the government's budget constraint.

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- the aggregate resource constraint,
- an **implementability condition** for consistency with household optimization,
- the government's budget constraint.

Lemma

The implementability condition on the planner's allocation choice is:

$$\sum_{t=0}^{\infty} \beta^t \left[u_c(t) c_t - u_{L_M}(t) L_{M,t} - u_{L_T}(t) L_{T,t} \right] = u_c(0) R_0 k_0.$$

Optimal Tax Collection

At an interior steady state, the optimal tax collection choice satisfies:

$$\frac{1 - \beta(1 - \delta_G)}{G_{L_C}(A^C, L_C) L_M} = \underbrace{\frac{F_G + 1 - \delta_G}{(F_k + 1 - \delta)(1 - \omega)} - 1}_{\text{net return of public capital relative to private capital}} + \underbrace{\beta \left(\frac{1 - \eta}{\eta} \right) F_{L_M} G L_C}_{\text{net benefit from wage increases - personnel pay}} \quad (4)$$

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Economic Interpretation:

- A high F_k (marginal return on private capital) favors private investment over public hiring.
- A high F_G (marginal return on public capital) enhances the attractiveness of public sector spending.
- A larger net marginal wage increase, net of tax personnel pay, increases the government's incentive to hire tax collectors

All endogenous objects.

4. Calibration and Counterfactual Effects of Tax Productivity Shocks

4.1 Calibration

Functional Forms

- We set the following functional forms:

[diminishing marginal products] $G(A^C, L_C) = A^C L_C^\eta,$

[Barro-Baxter-King complementary, [WBLRGM](#)] $F_M(k^G, k, L_M) = A_M(\theta k^G)^\zeta k^\mu L_M^{1-\mu},$

[linear traditional sector] $F_T(L_T) = A_T L_T,$

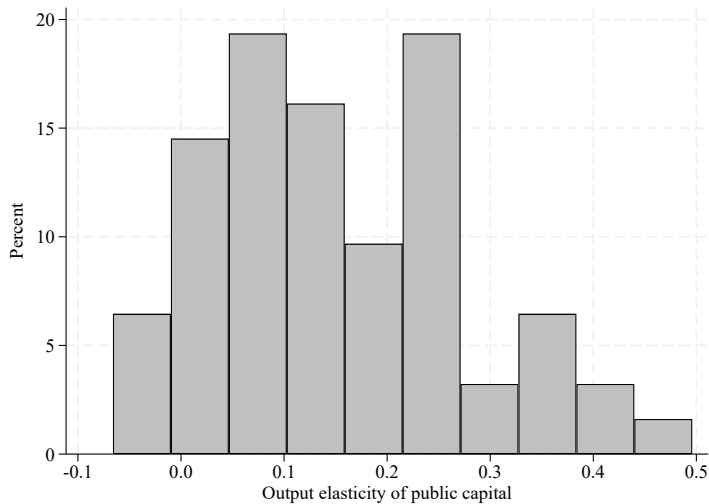
[CRRA] $u(c) = \frac{c^{1-\gamma}}{1-\gamma},$

[isoelastic disutility of labor] $u(L) = \frac{(L)^{1/\epsilon}}{1+1/\epsilon}.$

Three New Model Parameters

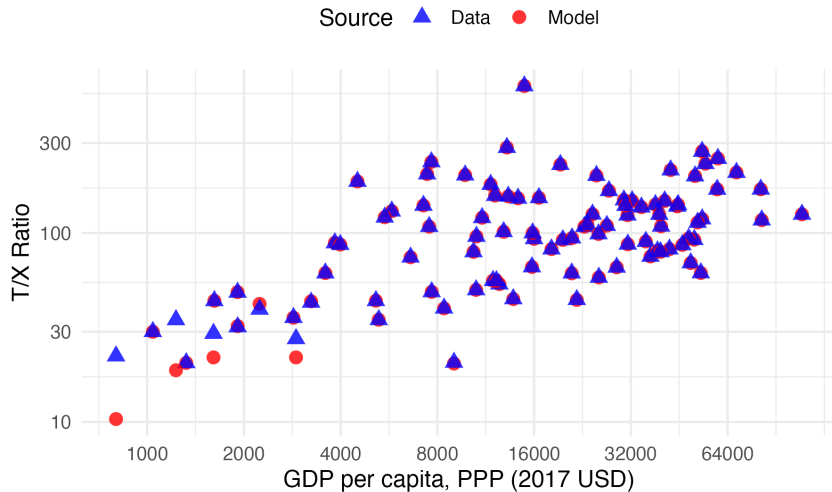
- Isoelastic tax collection: $\eta \approx \frac{\text{Marginal Product of Tax Collection}}{\text{Average Product of Tax Collection}}$, we set $\eta = 0.05$
 - LI: DJLT(2025) Ghana RCT, $d(T/X) = 1.56, T/X = 42, \implies \eta = 0.04$
 - MI: BSHK(2019), Costa Rica, $d(T/X) = 4, T/X = 82, \implies \eta = 0.05$
 - HI: BHSS(2025), US, $d(T/X) = 12.8, T/X = 260, \implies \eta = 0.05$
- ζ output elasticity of public capital, Bom and Ligthart (2014), Ramey (2021), wide range
 - We set a conservative value of $\zeta = 0.17$ from WB and $\zeta = 0.42$ from Ramey (2021)
- $\theta \leq 1$ WB's Infrastructure Efficiency Index (IEI); Devadas and Penningsv(2018)
 - We set $\theta_{low} = 0.58, \theta_{middle} = 0.74, \theta_{high} = 0.84$

Output Elasticity of Public Capital from Meta-Analysis

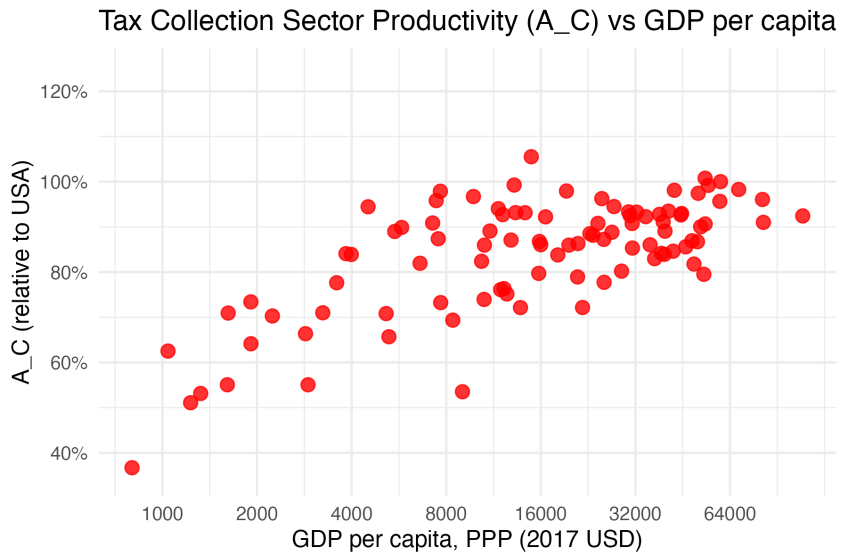


Targeted T/X

Tax-to-Expenditure Ratio (T/X): Model vs Data

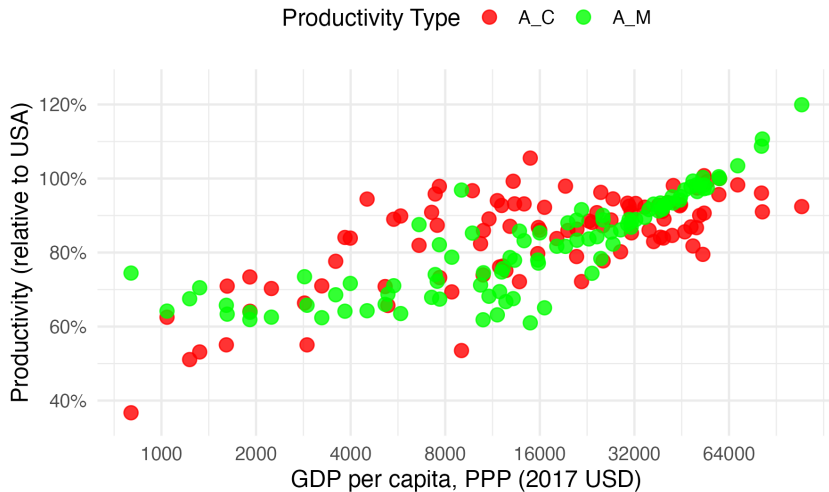


Inferred Tax Productivity Relative to US



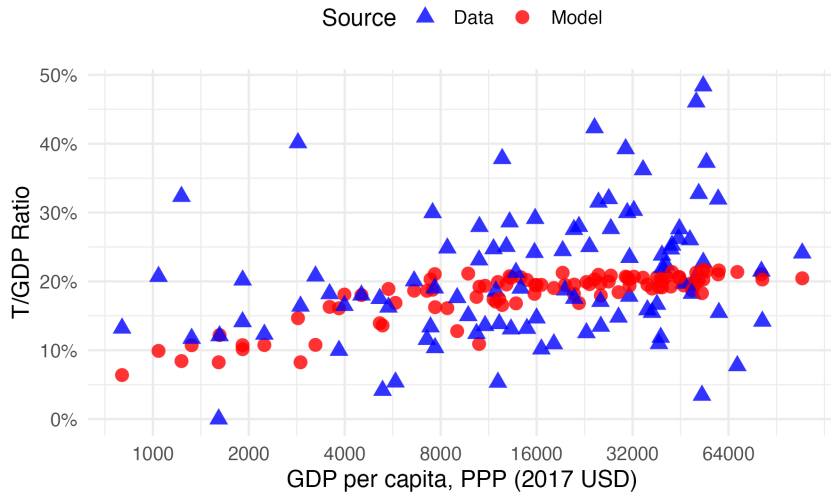
Variation in TFP vs. Tax Productivity

Productivity (A_C and A_M) vs GDP per capita



T/GDP and other Government Expenses

Tax-to-GDP Ratio (T/GDP): Model vs Data



4.2 Counterfactual Effects of Tax Productivity Shocks

Experiment: Raise Tax Productivity (A^C) of Ghana



See also Jawara, Knebelmann, Levine, Ndiaye, and Pouliquen (2025) [In the field. The Gambia]

Impact of Increase in Tax Productivity: Conservative Estimates

Table 1: Counterfactual Analysis Results for Ghana

ΔA_C	ΔGDP	$\Delta \text{Hulten}_{\text{im}}$	$\Delta \text{Hulten}_{\text{sm}}$	Δk	Δk_G	L_M/L	L_C/L	T/X
0	-	-	-	-	-	0.45	0.0014	31
10	1.1	0.01	0.31	2.4	12.3	0.45	0.0012	41
20	2.1	0.02	0.61	4.3	24.4	0.45	0.0009	54
50	4.4	0.05	1.53	8.6	57.2	0.46	0.0004	155

Columns with Δ indicate percentage changes in A_C , GDP, Hulten predictor, capital (k), public capital (k_G), and tax revenue. Variables L_M/L , L_C/L and T/X show absolute levels. Baseline parameters are $\beta = 0.96$, $\mu = 0.33$, $\delta = 0.06$, $\delta_G = 0.06$, $\gamma = 2.00$, $\epsilon = 0.50$, $\eta = 0.05$, $\zeta = 0.17$, $\theta = 0.58$, $A_M = 0.830$, $A_C = 0.130$, with a Domar weight of 0.00096942 (for inputs method) and tax/GDP ratio (for sales method) of 0.0305.

Impact of Increase in Tax Productivity: Ramey (2021) Estimates

Table 2: Counterfactual Analysis Results for Ghana

ΔA_C	ΔGDP	$\Delta \text{Hulten}_{\text{im}}$	$\Delta \text{Hulten}_{\text{sm}}$	Δk	Δk_G	L_M/L	L_C/L	T/X
0	-	-	-	-	-	0.43	0.0036	32
10	10.9	0.02	0.78	25.1	38.6	0.45	0.0030	43
20	20.4	0.05	1.55	45.6	76.6	0.46	0.0025	59
50	41.1	0.12	3.89	85.3	175.7	0.49	0.0008	221

Columns with Δ indicate percentage changes in A_C , GDP, Hulten predictor, capital (k), public capital (k_G), and tax revenue. Variables L_M/L , L_C/L and T/X show absolute levels. Baseline parameters are $\beta = 0.96$, $\mu = 0.42$, $\delta = 0.06$, $\delta_G = 0.06$, $\gamma = 2.00$, $\epsilon = 0.50$, $\eta = 0.05$, $\zeta = 0.39$, $\theta = 0.58$, $A_M = 0.585$, $A_C = 0.348$, with a Domar weight of 0.00241827 (for inputs method) and tax/GDP ratio (for sales method) of 0.0777.

Effect of Corruption

Table 3: Counterfactual Analysis Results for Change in θ

	θ	ΔGDP	Δk	Δk_G	L_M/L	L_C/L	T/X
θ_{low}	0.58	-	-	-	0.43	0.0036	32
θ_{middle}	0.74	48.5	118.9	120.0	0.51	0.0042	33
θ_{high}	0.84	76.8	184.6	186.0	0.55	0.0045	33

Columns with Δ indicate percentage changes in A_C , GDP, Hulten predictor, capital (k), public capital (k_G), and tax revenue. Variables L_M/L , L_C/L and T/X show absolute levels. Baseline parameters are $\beta = 0.96$, $\mu = 0.42$, $\delta = 0.06$, $\delta_G = 0.06$, $\gamma = 2.00$, $\epsilon = 0.50$, $\eta = 0.05$, $\zeta = 0.39$, $\theta = 0.58$, $A_M = 0.585$, $A_C = 0.348$, with a Domar weight of 0.00241827 (for inputs method) and tax/GDP ratio (for sales method) of 0.0777.

Conclusion

We used new data on tax collection over 100 countries and a workhorse Ramsey model with tax collection to demonstrate:

1. **Tax collection capacity increases with GDP per capita**
2. **Decreasing returns in tax collection can generate differences in long-run GDP**
3. **We provide Ramsey model (& novel formulas) for optimal state capacity**
4. **Quantitatively, GDP gains from tax productivity shocks are an order of magnitude larger than predicted by sectoral share accounting ("Hulten").**

Future work: Robustness, growth decomposition and accounting, transition dynamics (nothing special here), etc.

Empirical Findings: Tax Collections / Expenditures

Measuring Tax Collections/ Expenditure

- Unit of Observation: tax administration

Measuring Tax Collection / Expenditure

- Unit of Observation: tax administration



Measuring Tax Collection / Expenditure

- Data Source
 - National tax authority documents
 - IMF Revenue Administration Fiscal Information Tool
- 100 countries, 625 country-year observations
- T = Tax Collections including PIT, CT, VAT, sales, and customs taxes
- X = Official tax collection authority budget
- Tax collection over collection expenditure, T/X
 1. Cross-country comparisons and regressions
 2. Historical Panel Regressions
 3. Sub-national states regressions

Takeaways

- **T/X positively correlated with GDP per capita**
 1. across countries, robust to tax admin purview & rates
 2. over time
 3. across subnational states

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- **T/X positively correlated with GDP per capita**
 1. across countries, robust to tax admin purview & rates
 2. over time
 3. across subnational states
- Suggestive evidence, but need **structural model** to:
 - Formalize notion of **tax productivity**
 - Simulate effects of counterfactually raising tax productivity

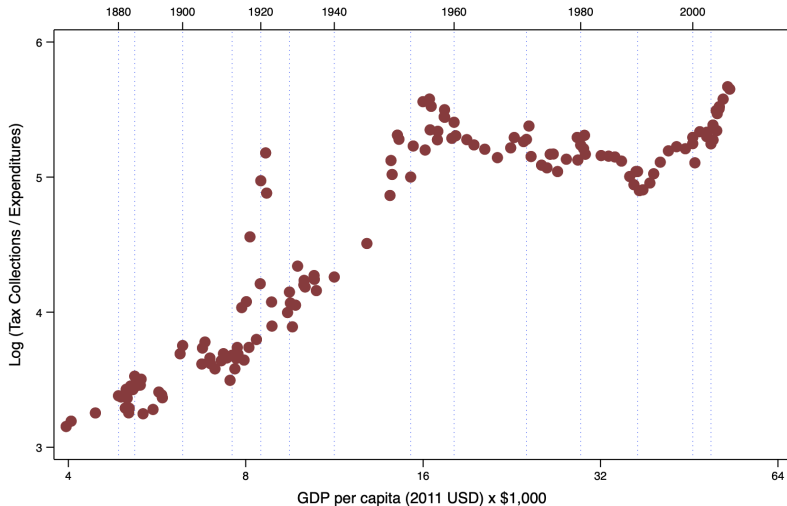
1. Descriptive Cross-Country Regressions: T/X and GDP Per Capita

Outcome	(1) Log T/X	(2) Log T/X	(3) Log T/X	(4) Log T/X	(5) Log T/X	(6) Log T/X	(7) Log T/X	(8) Log T/X
Log GDPpc	0.248*** (0.0389)	0.654*** (0.142)	0.225*** (0.061)	0.221*** (0.0586)	0.249*** (0.0395)	0.260*** (0.0398)	0.218*** (0.0484)	0.241*** (0.0399)
<i>N</i>	100	35	65	99	97	90	73	98
<i>R</i> ²	0.293	0.390	0.174	0.321	0.296	0.326	0.222	0.276
Modif. to column (1)		Only low, lower-middle inc. countries	Only upper- middle, high inc. countries	Cont. for PIT, VAT, CIT, tariff	Excl. if no VAT or sales tax	Excl. if no customs	Excl. if include soc. sec.	Excl. if resp for pop registry

[Statutory Tax Rates](#)

[Back to Main](#)

2. Tax Collection / Expenditure in the US: 1867 to 2017



2. Descriptive Panel Regressions over 16 Countries

	(1) Log (T/X)	(2) Log (T/X) Exclude if not responsible for customs duties	(3) Log (T/X) Exclude if responsible for social security contributions	(4) Log (T/X) Exclude if changes in responsibilities
Log Real GDP pc	0.637*** (0.175)	0.636*** (0.197)	0.684*** (0.163)	0.644*** (0.193)
Observations	625	546	519	578
R-squared	0.563	0.541	0.574	0.541
Country FE	Yes	Yes	Yes	Yes

[Back to Main](#)

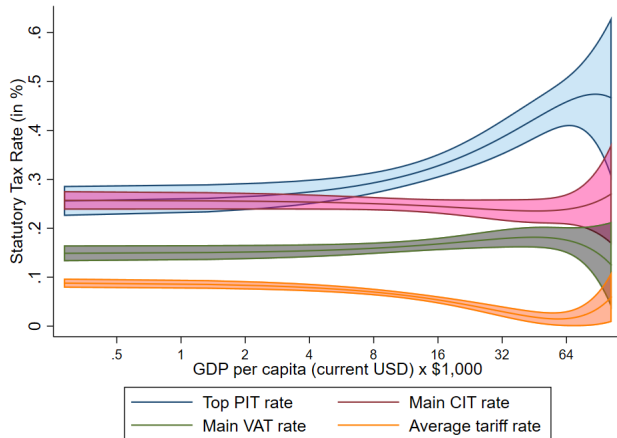
3. Sub-National States Regressions

Outcome	(1) Log T/X	(2) Log T/X
Log GDP percap	0.567*** (0.143)	0.278*** (0.0606)
Sample	US states (1993-2017)	Indian states (1950-2017)
Number of states	50	28
State FE	Yes	Yes
N	850	964
R^2	0.719	0.777

(1) US states (1993-2016), (2) Indian states (1950-2017)

[Back to Main](#)

Statutory Tax Rates in the Cross-Section of Countries



Notes: This figure plots the statutory tax rates against GDP per capita in the sample of 100 countries. The local polynomial for each type of tax rate is plotted, together with the 95 percent confidence interval.

Steady State

A steady state is determined by a system of 9 equations in 9 unknowns

$(w_M, w_T, r, L_M, L_T, L_C, k, k^G, c)$:

(i) Optimal collection:
$$\frac{1 - \beta(1 - \delta_G)}{G_{L_C}(A^C, L_C) L_M} = \beta \left[F_G + (1 - \delta_G) \right] - 1 + \beta \left(\frac{1 - \eta}{\eta} \right) F_{L_M C}$$

(ii) Private capital:
$$1 = \beta \left(F_k + 1 - \delta \right) + \beta \frac{1 - \eta}{\eta} F_{L_M, k} L_C,$$

(iii) Public capital:
$$k_G = \frac{1}{\delta_G} G(A^C, L_C) w_M L_M,$$

(iv) Return on private capital:
$$r = F_{M, k},$$

(v) Modern sector wage:
$$w_M = F_{M, L},$$

(vi) Traditional sector wage:
$$w_T = F_{T, L},$$

(vii) Modern labor supply:
$$u_{L_M} = u_c w_M (1 - G(A^C, L_C)),$$

(viii) Traditional labor supply:
$$u_{L_T} = u_c w_T,$$

(ix) Aggregate resource constraint:
$$c + k + w_M L_C + k_G = (1 + r - \delta)k + w_M L_M + A^T L_T.$$

Baseline Steady State

Variable	Parameter	Value
Modern labor	L_M	0.636
Traditional labor	L_T	0.773
Tax collectors	L_C	0.00192
Private capital	k	2.396
Public capital	k_G	0.755
GDP (output)	Y	1.484
Tax revenue	T	0.045
Average tax rate (%)	$\bar{\tau}$	9.51
Tax/GDP (%)	T/Y	3.05
Modern-sector share (%)	Y_M/Y	47.90

[Back to Main](#)

Transition Dynamics

Lemma

Assume that household preferences are separable in consumption and labor, with CRRA utility in consumption, and that the tax production function is linear in personnel $G(A^C, L_C) = A^C L_C$. Then an interior steady state exists and the optimal personnel choice over the transition (for $t > 0$) satisfies:

$$\frac{(F_G(t+1) - \delta_G) - (F_k(t+1) - \delta)}{F_k(t+1) + 1 - \delta} = \frac{1}{A^C} \left(\frac{1}{L_{M,t}} - \frac{\beta(1 - \delta_G)}{L_{M,t+1}} \right).$$

Higher tax productivity A^C leads to higher levels of public capital (lower F_G). [Back to Main](#)